

APPLIED PROBLEMS

On Mobile Pose Estimation and Action Recognition Design and Implementation

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Received Month DD, 20XX; revised Month DD, 20XX; accepted Month DD, 20XX

Abstract—Human Pose Estimation (PE, tracking body pose on-the-go) is a computer vision-based technology that identifies and controls specific points on the human body. These points represent our joints and special points over the body determining the sizes, distances, angle of flexion, and type of the motion. Knowing this in a specific exercise is the basis of work for rehabilitation and physiotherapy, fitness and self-coaching, augmented reality, animation and gaming, robot management, surveillance and human activity analysis. Implementing such capabilities may use special suits or sensor arrays to achieve the best result, but massive use of PE is related to devices that many users own—namely smartphones, smartwatches, and earbuds. The body pose estimation system starts with capturing the initial data. In dealing with motion detection, it is necessary to analyze a sequence of images rather than a still photo. Different software modules are responsible for tracking 2D key points, creating a body representation, and converting it into a 3D space. Action recognition on the other hand is a way to analyze the sequence of estimated pose data with the aim to categorize sequence under the classes. It is widely used in various fields. One of the widely known use cases is analysis and detection of potential attacks of illegal action using video from the surveillance cameras. Another use case involves analysis of the sequence of pose with the aim of creating a virtual coaching environment. Specifically, our research will target this challenging issue and aim to create this environment for mobile devices. We will describe some of the solutions that are suitable for effectively pose estimation and action recognition on mobile devices. We will show how lightweight models based on Convolution Neural Networks can be used to efficiently solve pose estimation issue and address action recognition problem with the Dynamic Time Warping algorithm.

Keywords: pose estimation, deep learning, dynamic time warping, convolution neural networks, action recognition

DOI: 10.1134/S1054661XXXXXXX

1. INTRODUCTION

Static and dynamic objects, together with the natural physical environment, constitute the infrastructure for the coexistence of the entire set of biological and mechanical systems [22]. The objects that move are mainly biological beings and systems controlled by them. In the

process of evolution, they are equipped with systems of observation, coordination and safe movement in the environment. We will be interested in the concept of pose estimation, with a global meaning of determining the absolute or relative position of an object or its elements in the environment [2]. This ability is inherent in biological beings and much research and work have been conducted to adopt a similar technique for artificial mechanical devices and systems. One of the major areas of research belongs to the autonomous cars industry with the goal of creating self-driving cars. In principle, we are talking about automating the visual sense of the situation through a system of video surveillance and possibly, sound, and traffic control on this basis. Other tasks from our field of interest are the task of drone self-guidance on a map without GPS, the task of detecting and tracking objects observed by radar, and of course, our main task of controlling the human body as a compound of complex critical functions [4, 9, 22, 38]. Specifically, this task is crucial in controlling robots for the work in natural and man-made disasters, work in outer space. The other major use cases include the tasks of organizing training and evaluating functions of specific exercises, which are critical at sports competitions, dance competitions, in the training of soldiers, rescuers, etc. [44, 45].

The two main components of video interpretation and automation systems are surveillance systems and data analysis systems. Usually, observations are made by one or many cameras placed at varying angles. The observed scene can also be specially prepared or an arbitrary environment can be used [6, 10]. According to the captured data, its analysis systems also differ, some start from standard image analyzes, which can use elements of image annotation, others can be trained without them, as conventional neural networks do [39, 41]. Success also depends on the specification of the task and the structure of the recorded data.

In this paper, we will not consider arbitrarily, but only focus on those moving objects that have a skeletal structure with hinges and a motor mechanism [3]. They are people and animals, as well as mechanical robots and devices. Each class that we study must be described with the structure, components, their properties and validation, etc. Input data that we consider for this task, can be direct raw images or their sequences [2, 6, 10, 12, 27] from one or several positions. Annotations [21] will also be used and will introduce additional information, developed both by man and by technical means. In the context of the previous annotated example, the utilization of machine learning systems can be explored alongside direct geometric or kinetic analysis of motion [14, 18, 25]. Typically, the optimal outcomes are achieved through the combined application of geometric analysis and artificial intelligence. The pivotal constituents of the scientific and technical systems under consideration encompass mathematical, computational, and instrumental components. These components consist of observation subsystems, interfaces, analysis modules, and decision-making processes. The precision of detection and interpretation is heavily reliant on the quality of the technical equipment in use. Moreover, the extent of the system's capabilities is also contingent on this equipment.

To elaborate the configuration of the system can vary widely [23, 27]. It can range from an elaborate setup in a dedicated studio replete with multiple cameras and specialized recording devices, to the integration of additional sensors on observed objects. Conversely, it could involve nothing more than the commonplace pairing of a regular smartphone with a projector. In the former professional scenario, tasks necessitating exacting precision can be effectively tackled. In contrast, the latter case is tailored to address the needs of a broader user base.

The focus of our research resides in the latter category. It will be demonstrated that even with modest technical support, it is feasible to resolve these challenges with the requisite accuracy [13, 21]. This is achievable due to the incorporation of modern smartphones, tablets, and similar devices, which come equipped with the requisite engineering for data capture and analysis [31].

The following paper is constructed with the following structure. In the first part we will address the issue of pose estimation from a single image. We will cover some of the challenges, the datasets that we have used and the performance of different models.

The second part of our study will cover an efficient way for action recognition from the sequence of 3D body landmarks. Specially we will use the dynamic time warping algorithm, which requires minimal computation resources for invocation and is relatively simple in terms of implementation and invocation on mobile devices.

2. DESIGN OF THE POSE ESTIMATION SYSTEM

The problem of 3D pose estimation is challenging due to the large variety of human poses and body proportions, occlusion, and the diversity of scenes, angles of observation, and backgrounds against which humans are depicted. Analysis from a single RGB image is particularly challenging due to the lack of depth information, which leads to geometric ambiguity and occlusion. The limited coverage of current datasets is a major obstacle to training models for human visual analysis, as 2D datasets like LSP [20] and MPI-II [1] offer only approximate 2D joint location ground truth and lack human segmentation or body part labeling information, while 3D datasets like HumanEva [36] and Human3.6M [19] offer accurate 2D and 3D reconstructions but are limited in terms of background and clothing variability. The proposed solution by [35] is a deep learning model that can fully automatically sense humans at multiple levels of detail, including figure-ground segmentation, body-part labeling at the pixel level, and 2D and 3D pose estimation, using multi-task loss functions at different processing stages. The model uses multitask loss functions at different recursive processing stages, including human body joint detection and 2D pose estimation, semantic body part segmentation, and 3D reconstruction, to effectively train on multiple datasets that would otherwise cover only some of the model components. In experiments using representative 2D and 3D datasets like LSP, HumanEva, and Human3.6M, Popa et al. [28] demonstrate that the model achieves state-of-the-art results for both semantic body part segmentation and 3D pose estimation.

2.1. Introduction to Skeletal Structure and Its Components

The main component of the skeleton of biological beings is bones. The number and structure of bones in the same biological individuals are not constant. A person has 360 bones, but there is a difference associated with sex, and this number changes over the years when some components are fused into one. Along with this, there is a reference table indicating all shapes and sizes and their possible defects [3].

The second important component of the skeleton is the mechanism of connection and interaction of bones, which is called a joint. There is a developed system of joint compounds and, of course, a reference system for the types and joints and functionality they are responsible for. At the technical level, types are distinguished such as synovial, cartilaginous and fibrous joints. Together they represented the structure shown in Fig. 1.

2.2. What Do Joints Do?

Joints connect bones. They provide stability to the skeleton, and allow movement. There are different types of joints.

2.2.1. Synovial joints

Joints in the arms and legs are synovial joints. The ends of the bones are covered with cartilage and separated by the joint cavity, which is filled with a thick gel called synovial fluid. Synovial

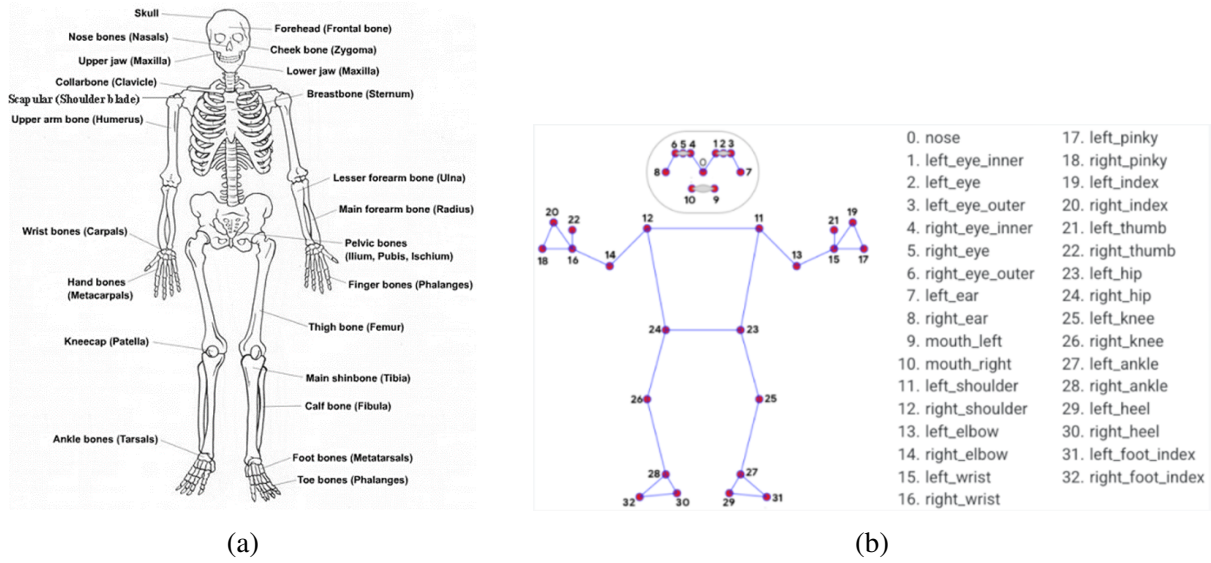


Fig. 1. Major bones of the human skeleton (a), and the corresponding topology of 33 body landmarks used by the estimator (b), in which every named point is one of the joints or surface points that the model has to return

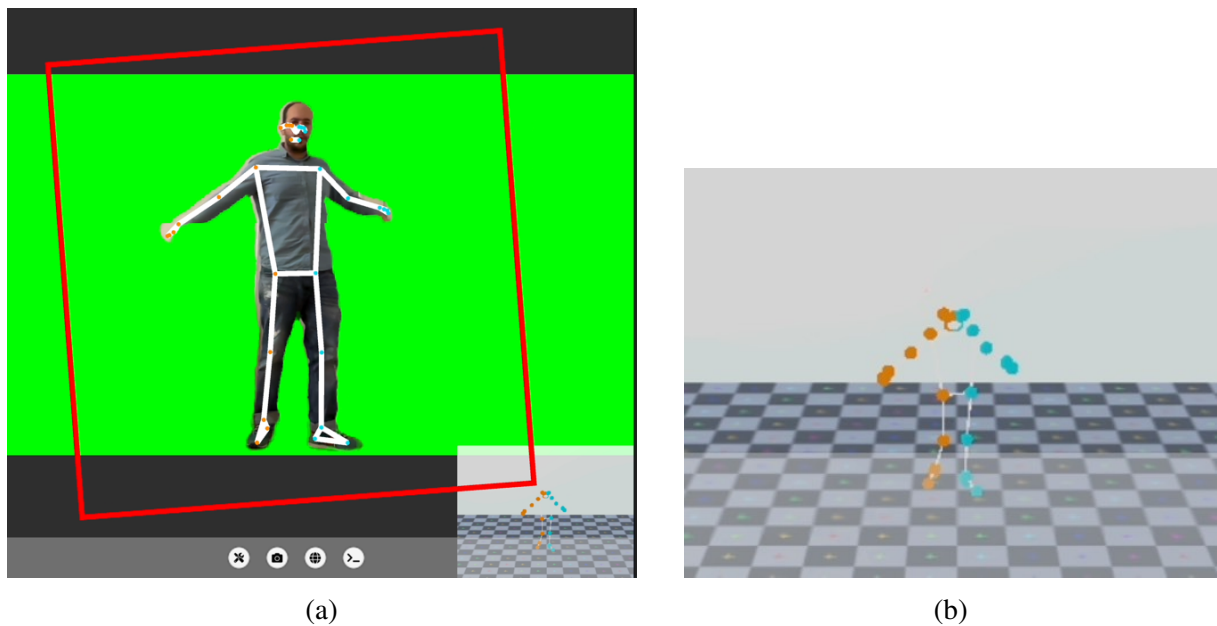


Fig. 2. Render of the output from a single frame of the video with body movements. The frame shows the rectangle of the detected human body, the segmented region of the body in green, and the 2D landmarks rendered on top of the human body (a), together with the window of 3D landmarks in 3D space (b)

fluid helps to lubricate the cartilage and provides nourishment to it. Ligaments stretch across the joint, connecting one bone to another and help to stabilize the joint so it can only move in certain directions.

2.2.2. Cartilaginous joints

Joints in the spine and pelvis and the joints between the ribs and the sternum are cartilaginous joints—they provide more stability but not as much movement. The bones are connected by cartilage in this type of joint.

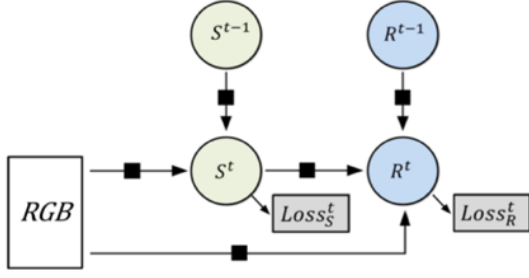


Fig. 3. Stage t of our architecture for recognition and reconstruction

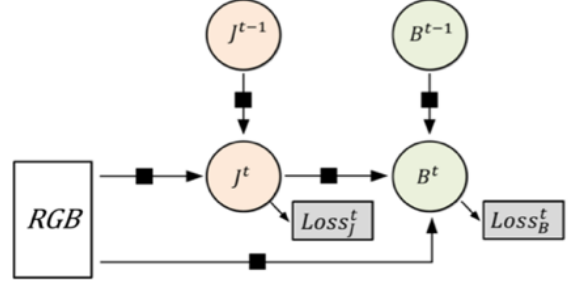


Fig. 4. Stage t of our semantic task, including 2D joint detection (J), and labeling of the body parts (B)

2.2.3. Fibrous joints

Fibrous joints allow no movement—only stability. They are held together by fibrous connective tissue and located on the skull.

2.3. Deep Convolution Neural Networks for 2D and 3D Pose Landmarks Reconstruction from Monocular Images

Most recent and most efficient methods for 3D landmarks reconstruction are proposing to handle this problem with a multistage architecture, the first stage of which includes 2D pose estimation followed by figure segmentation and finally 3D pose reconstruction. The recurrent feed-forward processing architecture used by [28] has achieved state of the art results to the current problem. The concept proposed by Popa et al. includes semantic processing stages denoted by S^t and R^t . Specifically, S^t consists of 2 subtasks, J^t which is designed for 2D pose estimation and B^t for body parts labeling and figure ground segmentation. Stage R^t , which takes the output of S^t as an input, handles 3D reconstruction. We will discuss the work of each stage and its subtasks next in the order of its appearance.

2.3.1. 2D human body joint detection

Pose inference machines introduced a new structure for pose estimation, suggesting a multiple-level detection approach with an increase over stages of the number of pose landmarks and propagating detection belief maps to the following stages [29]. Methods suggested by Ramakrishna et al. are further improved by replacing feature detection, training, and prediction algorithms with convolution neural networks [47].

Given an RGB image $I \in \mathbb{R}^{w \times h \times 3}$ we seek to correctly predict the locations of N_J anatomically defined human body joints $p_k \in Z \subset \mathbb{R}^2$, with $k \in \{1 \dots N_J\}$. At each stage $t \in \{1 \dots T\}$, where T is the total number of stages, the network outputs belief maps $J^t \in \mathbb{R}^{w \times h \times N_J}$.

A pose machine consists of a sequence of multi-class predictors, $g_t(\bullet)$, that are trained to predict the location of each part in each level of the hierarchy [29]. For each stage there are 3 levels defined. The first one detects the body, the second level addresses limb detection and the final third level detects the region around pose landmarks. At each level l and stage $t \in \{1 \dots T\}$, the classifiers g_t predict beliefs for assigning a location to each part $Y_p = z$, $\forall z \in Z$, based on features extracted from the image at the location z denoted by $x_z \in \mathbb{R}^d$ and contextual information from the preceding classifier in the neighborhood around each Y_p in stage t . The output of the first stage classifier therefore can be denoted as

$$g_1(x_z) \rightarrow \{b_1^p(Y_p = z)\}_{p \in \{0 \dots P\}}, \quad (1)$$

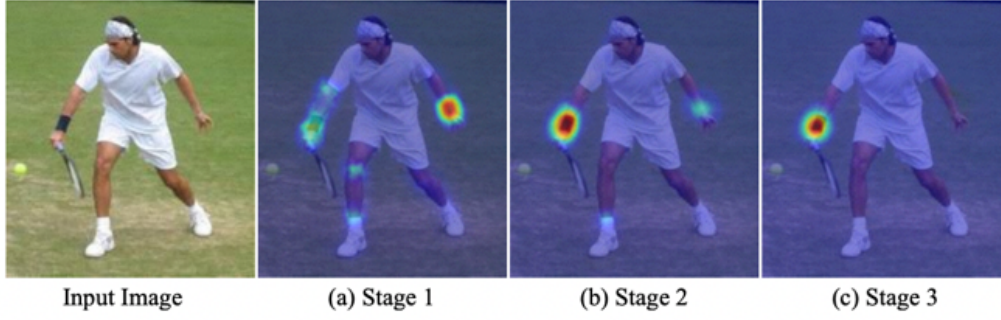


Fig. 5. A Convolutional Pose Machine

where $b_1^p(Y_p = z)$ is the score predicted by the classifier g_1 for assigning the p th part in the first stage at image location z . In subsequent stages, the classifier predicts a belief map by assigning a location to each part $Y_p = z, \forall z \in Z$, based on features of the image data $x_z^t \in \mathbb{R}^d$ again, and contextual information from the preceding classifier in the neighborhood around each Y_p .

The design of the convolution pose machine, Fig. 5, is based on an architecture similar to [29] with replacement of feature detection, training and prediction functionality with fully CNNs [24]. Receptive fields used at the CNN increase over stages. This is done to supply contextual information to the convolution layers at subsequent stages and allow the learning of long-range correlations between body parts, therefore increasing detection accuracy.

The first stage of the CPM generates belief maps from a CNN that examined the image locally with a small receptive field. The CNN at this stage consists of 7 convolution and three pooling layers producing x features. At stages $t \geq 2$ (where t is the stage number) multiple convolutional layers are used to achieve a large receptive field on the $8\times$ downscaled heatmap. Through the series of experimental trials on input images normalized to a size of 304×304 with the FLIC dataset [33] it has further been proven that accuracy improves with an increase of the size of receptive fields. It is found that the best performing architecture crops and scales the image into a size of 368×368 pixels for better precision. Further, the receptive field of the second stage outputs the belief maps of the first stage into a size of 31×31 . The classification function c_j^t is used next for body joint belief maps J^t production. c_j^t consists of a series of five convolutional layers, the first three of the form $(11 \times 11 \times 128)$, followed by a $(1 \times 1 \times 128)$ convolution and a final $(1 \times 1 \times N_J)$ convolution that outputs J^t .

The loss function at each stage L_j^t minimizes the squared Euclidean distance between the predicted and ground truth belief maps, J^t and J^* . Standard stochastic gradient descent is used to jointly train all the T stages in the network.

In practice, this component of the model can be trained with data from both 2D and 3D datasets like LSP [20], where the ground truth is obtained manually, or HumanEva [36] and Human3.6M [19] where the ground truth is obtained automatically based on anatomical markers:

$$L_j^t = \sum_{k=1}^{N_J} \sum_{z \in Z} \|J^t(z, k) - J^*(z, k)\|_2^2. \quad (2)$$

2.3.2. Semantic body part segmentation

In semantic body part segmentation (body part labeling) we assign each image location $(u, v) \in Z \subset \mathbb{R}^2$ one of N_B anatomical body part labels (including an additional label for background), b_l , where $l \in \{1 \dots N_B\}$. At each stage t , the network predicts, for each pixel location, the probability of each body part being present, $B^t \in \mathbb{R}^{w \times h \times N_B}$ [24].

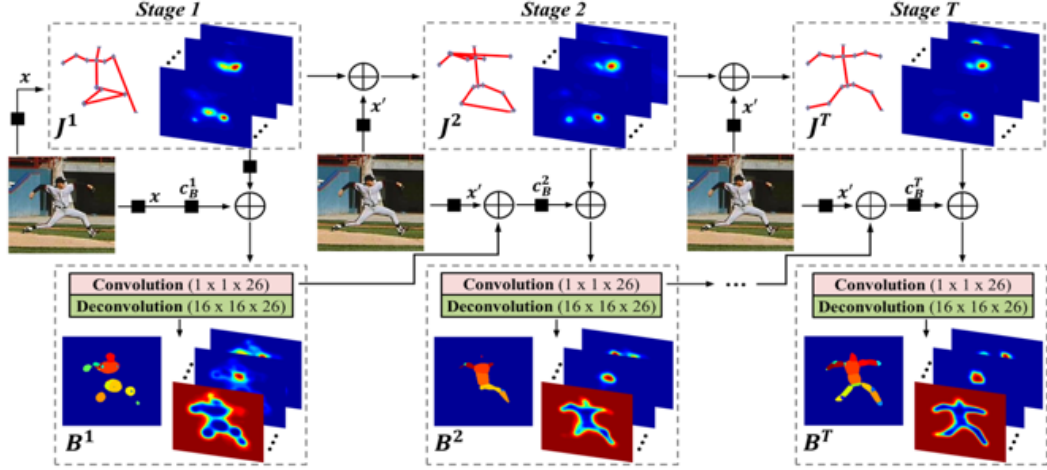


Fig. 6. Multitask multistage 3D reconstruction module R^t , combines 3D processing with information from semantic modules, S^t

The loss function used changes from a squared Euclidean loss to a multinomial logistic loss:

$$L_B^t = -\frac{1}{|Z|} \sum_{z \in Z} \log(B_{z, B_z^*}^t), \quad (3)$$

where B_z^* is the ground truth label for each image location $z = (u, v)$. Validation is done with the standard metric of mean pixel intersection over union.

In the first stage of processing, the model uses a series of convolution and pooling layers (with parameters tied from the first part) to extract features from the input image. This information was combined with 2D pose belief maps J^1 . These features are then used to make predictions about the current body labels B^1 . For each of the following stages, we also use the information present in the body labels at the previous stage, B^{t-1} , and rely on a series of four convolutional layers c_B^t that learn to combine inputs obtained by stacking image features x and B^{t-1} . The function c_B^t shares the same structure as the first four convolutions in c_J^t , but a classifier in the form of a $(1 \times 1 \times N_B)$ convolution is applied after the fusion with the current 2D pose belief maps J^t to obtain semantic probability maps B^t . Finally, an additional deconvolution layer of size $16 \times 16 \times N_B$ is used [24], such that the loss can be computed at the full resolution of the input image I , Fig. 6.

Training data was obtained from Human3.6M and also approximately generated from the LSP dataset. The Human3.6M is a large dataset of captured human poses performing daily activities. It is fully annotated and includes RGB data with ground truth masks for 2D body part labeling.

2.3.3. 3D pose reconstruction

The 3D reconstruction module suggested by [28] uses all gathered information to construct 3D landmarks. Specifically, 2D semantic components S^t , joint labeling J^t and segmentation map B^t . In order to obtain body reconstruction maps, feature maps D^t , trainable function c_D^t was defined similar to c_B^t .

The complete architecture can be summarized with the following steps: we first fuse S^t and D^t , then apply a series of one $(3 \times 3 \times 128)$ convolution, one $(3 \times 3 \times 64)$ convolution, one $(1 \times 1 \times 64)$ convolution, followed by a pooling layer (3×3) and a $(1 \times 1 \times 16)$ convolution. The output is concatenated with R^{t-1} and convolved by a $(1 \times 1 \times 16)$ kernel that learns to combine

Table 1. 3D mean joint position error on the Human80K dataset. Different components of the model of [28] are compared to [24]

Model	Avg. MPJPE (mm)
Model (J)	128.05
Model (D)	77.56
Model (J, B)	118.68
Model (J, D)	72.00
Model (J, B, D)	63.35

the two components, producing the estimate R^t . The feature maps are then transformed to the desired dimensionality of the 3D human body skeleton by means of a fully connected layer. The loss L_R^t is expressed as the mean per joint position error (MPJPE):

$$L_R^t = \sum_{i=1}^{N_R} \sqrt{\sum_{j=1}^3 (f(R^t, i, j) - R^*(i, j))^2 + \varepsilon^2}, \quad (4)$$

where R^* are the 3D ground truth human joint positions, $f(\cdot)$ is the fully connected layer applied to R^t and ε is a small constant that makes the loss differentiable. This loss component can be trained with data from HumanEva and Human3.6M. Although images at HumanEva and Human3.6M are captured at indoor laboratory settings, the use of a multitask loss makes the complete 2D and 3D model competitive also in the wild.

The complete multitask, multistage loss function of the model goes as follows:

$$L = \sum_{t=1}^T (L_J^t + L_B^t + L_R^t). \quad (5)$$

The loss allows us to conveniently train all the model component parameters, for different tasks, based on datasets where the annotations are challenging, or where annotations are missing entirely [48]. For the evaluation of the method by [28] the Leeds Sports Dataset (LSP), HumanEva and Human3.6M were used as they are both challenging datasets, containing complementary poses, and offering accurate 3D pose annotations. It is observed that the fully integrated system, $DMHS_R(J, B, D)$, achieves the lowest error of 63.35 mm, and that incorporating the body joint belief maps (J) and the body labeling probability maps (B) significantly reduces the error compared to a model based on feed-forward processing and inferring additional 2D joint position information. The results of the model are also compared to the state-of-the-art results on Human80K reported in a previous study, Table 1.

2.4. Conclusion of Pose Estimation

The proposed module is designed to estimate the figure-ground segmentation, detect human body joints, semantically identify body parts, and reconstruct the 2D and 3D pose of a person. It uses recurrent multitask loss functions at multiple stages of processing to effectively combine the strengths of different 2D and 3D datasets. The effectiveness of the system is demonstrated through ablation studies and evaluations of the effect of various types of training data on the multitask loss. The system is shown to achieve state-of-the-art results at all processing levels and to be competitive with state-of-the-art commercial RGB-D systems even in challenging real-world conditions.

3. DESIGN OF THE ACTION RECOGNITION SYSTEM

3.1. General Description

We use the popular HAR (Human Activity Recognition) dataset from UCI [31], which contains labeled time series. Specifically, an instance of this dataset is a person wearing a smartphone, which captures the linear acceleration and angle velocity while performing one of the following activities (WALKING, WALKING_UPSTAIRS, SITTING, WALKING_DOWNSTAIRS, STANDING, LAYING). Therefore, each observation is a 561-feature vector with time/frequency domain variables and a label describing the person's activity, and the goal is to build a model that accurately predicts the activity using the transformed feed from the smartphone.

3.2. Time Series Similarity Measures

3.2.1. Euclidean distance

One of the simplest similarity measures for time series is the Euclidean distance measure. Assume that both time sequences are of the same length n ; we can view each sequence as a point in n -dimensional Euclidean space, and define the dissimilarity between sequences as the familiar Euclidean distance. This measure is simple to understand and easy to compute, being the most widely used distance measure for similarity.

3.2.2. Longest common subsequence similarity

The longest common subsequence similarity measure is a variation of edit distance used in speech recognition and text pattern matching. The basic idea is to match two sequences by allowing some elements to be unmatched. The advantage of the LCSS method is that some elements may be unmatched or left out (e.g., outliers), whereas in Euclidean all elements from both sequences must be used, even the outliers.

3.2.3. Dynamic time warping

More flexible for some applications is the Dynamic Time Warping distance measure, that we use in [16, 32]. This is preferable when two sequences have approximately the same overall shapes, but these shapes do not line up in the X -axis. In order to find a similarity between such sequences we must “warp” the time axis of one sequence to achieve a better alignment.

Consider two gesture sequences (of possibly different lengths) to be compared against each other as two time series

$$X = (x_1, x_2, \dots, x_{t_1}, \dots, x_{T_1}), \quad Y = (y_1, y_2, \dots, y_{t_2}, \dots, y_{T_2}). \quad (6)$$

Using multivariate series, these two sequences form a much larger feature vector for comparison. Evidently, it is impossible to compute a distance metric between two vectors of unequal dimensions. A local cost measure is defined as

$$d : \mathcal{F} \times \mathcal{F} \rightarrow \mathbb{R}_{>0}, \quad x_{t_1}, y_{t_2} \in \mathcal{F}, \quad (7)$$

for $t_1 \in [1, T_1]$, $t_2 \in [1, T_2]$. By evaluating the cost matrix for all elements in X and Y , we obtain the matrix $C_{T_1 \times T_2}$. From this local cost matrix, we wish to obtain a correspondence mapping element in X to the elements in Y that will result in the lowest distance measure. We can define this mapping correspondence as

$$f = c(1), c(2), \dots, c(k), \dots, c(K), \quad c(k) = c(x_k, y_k). \quad (8)$$

The mapping function should follow the time sequence order of the respective gestures. Hence, we impose several conditions on the mapping function.

1. *Boundary conditions*: the starting and ending observation symbols are aligned to each other for both gestures, $c(1) = c(x_1, y_1)$ and $c(K) = c(x_{T_1}, y_{T_2})$.
2. *Monotonic condition*: the observation symbols are aligned in the order of time. This is intuitive as the order of observation signals in a gesture signal should not be reversed, $k_1 \leq k_2 \leq \dots \leq K$.
3. *Step size condition*: no observation symbols are to be skipped, $k_{i+1} - k_i \leq 1$.

In this way, we arrive at an overall cost function defined as

$$C(X, Y) = \sum_{k=1}^K c(k), \quad (9)$$

which gives an overall cost/distance between two gestures according to a warping path, as defined by the function f . Since the function $C(X, Y)$ denotes all possible warping paths between two gesture observation sequences X and Y , the dynamic time warping algorithm is to find the warping path, which gives the lowest cost/distance measure between the two gestures.

In our scenario, we apply dynamic programming principles to calculate the distance to each $c(k)$. We define D as an accumulated cost matrix:

1. initialize $D(1, 1) = d(x_1, y_1)$;
2. initialize $D(T_1, T_2)$ to an arbitrarily large number;
3. calculate

$$D(t_1, t_2) = \min\{D(t_1 - 1, t_2 - 1), D(t_1 - 1, t_2), D(t_1, t_2 - 1)\}. \quad (10)$$

3.3. Orientation Representation

We present orientation using quaternions. Quaternions are a compact and complete representation of rotations in 3D space compared with Euler angles. Quaternions are built from 4-dimension tuples (W, X, Y, Z) . In a quaternion representation of rotation, singularities are avoided, giving a more efficient and accurate representation of rotational transformations. A quaternion, which is of 4 dimensions, has a norm of 1, and is typically represented by one real dimension and three imaginary dimensions. The three imaginary dimensions, which are i , j , and k , are unit length and orthogonal to one another:

$$q = xi + yj + zk + w, \quad w^2 + x^2 + y^2 + z^2 = 1, \quad (11)$$

$$ij = k, \quad ji = -k, \quad ik = -j, \quad ki = j. \quad (12)$$

Quaternions (w, x, y, z) typically represent a rotation about the (x, y, z) axis by an angle of

$$\alpha = 2 \cos^{-1} w = 2 \sin^{-1} \sqrt{x^2 + y^2 + z^2}. \quad (13)$$

Although each series consisted of n (number of joints in the model) serialized quaternions, it will be split up into its individual quaternions for metric calculation. The final distance will be the sum of the distance between the n pairs of quaternions.

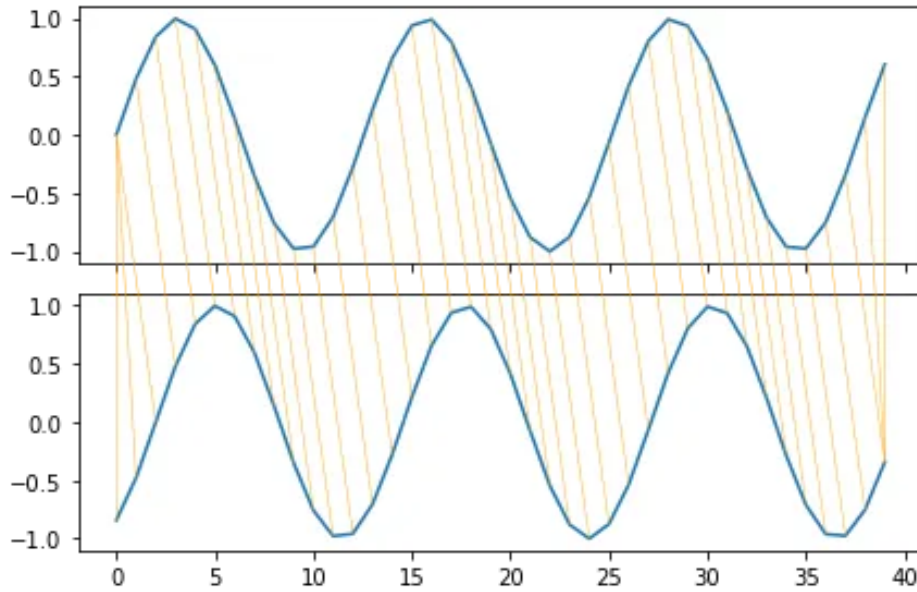


Fig. 7. Optimal warping distances between the 2 series

3.4. Recognition Procedure

In the simple case, we envision the use of a typical NN algorithm, where there is no specific learning phase. The system stores a list of multivariate time series of known activities and their corresponding labels in a database. When an unknown action is presented to the system, the system takes the unknown time series, and performs a sequential search with lower bounding DTW.

For the sake of implementation of dynamic time warping in Python, we will use the `dtw` package code, shown below. In this example, we will use 2 sine signals. One of the phases of the sine wave signals is shifted and DTW is used to assign a specific group from `signal1` to the second signal.

```
from dtw.distance import dtw
from dtw.distance import dtw_visualisation as dtwvis
import random
import numpy as np

x = np.arange(0, 20, .5)
s1 = np.sin(x)
s2 = np.sin(x - 1)
path = dtw.warping_path(s1, s2)
dtwvis.plot_warping(s1, s2, path)
distance = dtw.distance(s1, s2)
```

Figure 7 shows the optimal distances between all points of the 2 sine waves. We can also plot the dynamic programming matrix (or accumulated cost matrix), which shows all the warping paths. This is shown in Fig. 8.

Each cell in Fig. 8 is actually a number, representing the distance between the 2 respective data points being compared, one for each sequence. The darker the color, the lower the distance. After constructing the matrix, the optimal warping path is extracted (red line).

On the other hand, the time complexity is $O(M, N)$ where M, N are the lengths of the respective sequences—a quadratic cost. Considering that the sequences may be large (not un-

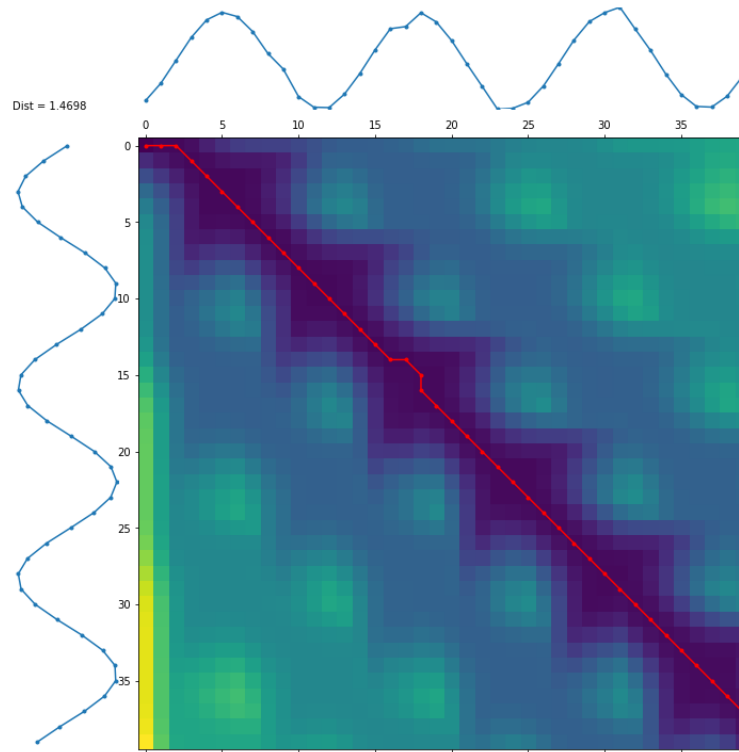


Fig. 8. Cost matrix. The red line indicates the optimal warping path

common in real-world examples) as well as the fact that KNN would still have to run afterwards, it is very likely that the model may take too long to get trained.

3.5. Use Cases and Results

Dynamic Time Warping emerges as a valuable tool for action recognition not only in the fitness industry but also across a spectrum of applications. Its scientific foundation and unique ability to align and compare time-series data make it a versatile technique to address the nuanced and dynamic nature of human movements [5, 7, 20, 26, 30, 33, 35, 42, 43, 46, 50].

In the fitness and virtual-coaching realm, DTW plays a pivotal role in exercise classification, repetition counting, form assessment, and customized training. This enables the trainers and individuals to personalize workout routines, monitor progress, and enhance overall fitness outcomes, Figs. 9 and 10.

In the animation and gaming industry, DTW facilitates the real-time recognition of human gestures and movements, enhancing user experiences through immersive interactions. Gamification of fitness routines becomes more engaging and effective when DTW is employed for action recognition and can be used for interaction with virtual environments.

In the realm of surveillance and human activity analysis, DTW contributes to identifying and classifying human movements in real-time video streams. This has applications in security and safety, where detecting anomalous activities or identifying specific actions in crowded environments is crucial.

While DTW offers numerous benefits, it does face computational challenges, particularly in real-time applications and large-scale datasets [11, 26, 39]. Future directions include the integration of machine learning techniques to enhance its accuracy and efficiency, and the development of hybrid approaches that combine DTW with other alignment techniques to mitigate computational demands.

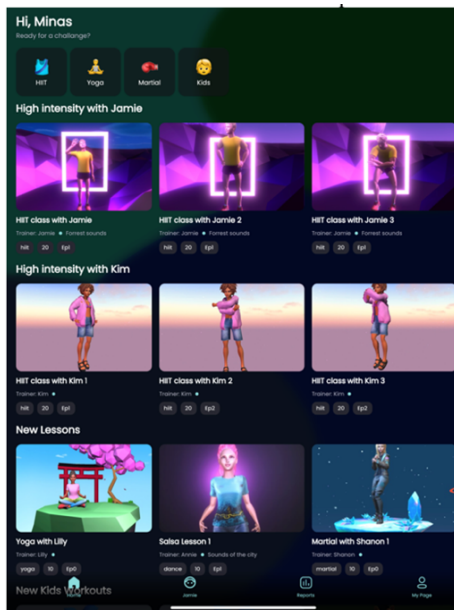


Fig. 9. The general menu of the system with basic use cases

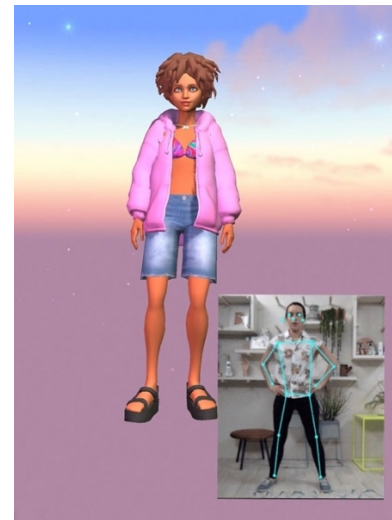


Fig. 10. AI-based workout application with a virtual coach (on the main screen) and a trainee (on the supplementary frame) with an estimated pose

In conclusion, Dynamic Time Warping's applicability spans across the fitness industry, self-coaching, animation and gaming, robot management, surveillance, and human activity analysis. Its scientific underpinnings make it a powerful tool for recognizing and analyzing complex time-series data, thereby contributing to advancements in various domains and ultimately enriching human experiences.

4. CONCLUSIONS

This paper shows how pose estimation and action recognition techniques have been integrated into mobile devices for the creation of performance efficient and visually pleasing software for virtual coaching. Specifically, we have shown that a stack of multiple convolution neural networks can be effectively used for real-time pose estimation in 3D space for mobile devices. Further we have fed sequential data of 3D reconstructed landmarks into the dynamic time warping algorithm in order to classify the performed action. The dynamic time warping algorithm that we have chosen as a main baseline for action recognition has shown impressive results during invocation on mobile devices with very limited computational resources and further has proved its efficiency for the problem of action recognition on mobile devices. For the sake of usability, we have developed a mobile application with a 3D humanoid virtual trainer character which is able to interact with the users and provide real-time insights about the performance of certain fitness exercises and movements.

FUNDING

The work is partially supported by grant No. 21T-1B314 of the Science Committee of the Ministry of Education, Science, Culture and Sports of the Republic of Armenia.

CONFLICT OF INTEREST

The process of writing and the content of the article does not give grounds for raising the issue of a conflict of interest.

COMPLIANCE WITH ETHICAL STANDARDS

This article is a completely original work of its author; it has not been published before and will not be sent to other publications until the PRIA editorial board decides not to accept it for publication.

REFERENCES

- [1] Andriluka, M., Pishchulin, L., Gehler, P., and Schiele, B., 2D human pose estimation: New benchmark and state of the art analysis, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2014, pp. 3686–3693.
- [2] Andriluka, M., Roth, S., and Schiele, B., Pictorial structures revisited: People detection and articulated pose estimation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2009, pp. 1014–1021.
- [3] Dantone, M., Gall, J., Leistner, C., and Van Gool, L., Human pose estimation using body parts dependent joint regressors, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 3041–3048.
- [4] Doherty, L., Pister, K.S.J., and El Ghaoui, L., Convex position estimation in wireless sensor networks, *Proc. IEEE INFOCOM 2001, 20th Annu. Joint Conf. of the IEEE Computer and Communications Society*, USA, 2001, vol. 3, pp. 1655–1663.
- [5] Duchi, J., Hazan, E., and Singer, Y., Adaptive subgradient methods for online learning and stochastic optimization, *Proc. 23rd Conf. on Learning Theory (COLT)*, 2010, pp. 257–269.
- [6] Eichner, M. and Ferrari, V., Better appearance models for pictorial structures, *Proc. British Machine Vision Conf. (BMVC)*, London, 2009.
- [7] Eichner, M., Marin-Jimenez, M., Zisserman, A., and Ferrari, V., Articulated human pose estimation and search in (almost) unconstrained still images, *Int. J. Comput. Vis.*, 2012, vol. 99, no. 2, pp. 190–214.
- [8] Everingham, M., Van Gool, L., Williams, C.K.I., Winn, J., and Zisserman, A., The PASCAL Visual Object Classes Challenge 2011 (VOC2011) Results, 2011. <http://www.pascal-network.org/challenges/VOC/voc2011/workshop/index.html>
- [9] Faisal, A., Majumder, S., Mondal, T., Cowan, D., Naseh, S., and Deen, M.J., Monitoring methods of human body joints: State-of-the-art and research challenges, *Sensors*, 2019, vol. 19, no. 11, p. 2629.
- [10] Felzenszwalb, P.F. and Huttenlocher, D.P., Pictorial structures for object recognition, *Int. J. Comput. Vis.*, 2005, vol. 61, no. 1, pp. 55–79.
- [11] Ferrari, V., Marin-Jimenez, M., and Zisserman, A., Progressive search space reduction for human pose estimation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2008, pp. 1–8.
- [12] Fischler, M.A. and Elschlager, R.A., The representation and matching of pictorial structures, *IEEE Trans. Comput.*, 1973, vol. C-22, no. 1, pp. 67–92.

- [13] Girshick, R., Donahue, J., Darrell, T., and Malik, J., Rich feature hierarchies for accurate object detection and semantic segmentation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2014, pp. 580–587.
- [14] Gkioxari, G., Arbeláez, P., Bourdev, L., and Malik, J., Articulated pose estimation using discriminative armlet classifiers, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 3342–3349.
- [15] Grishchenko, I., Bazarevsky, V., Zanfır, A., Bazavan, E.G., Zanfır, M., Yee, R., Raveendran, K., Zhdanovich, M., Grundmann, M., and Sminchisescu, C., BlazePose GHUM Holistic: Real-time 3D human landmarks and pose estimation, *arXiv preprint*, 2022, arXiv:2206.11678.
- [16] Huu, P.C., Le, Q.K., and Le, T.H., Human action recognition using dynamic time warping and voting algorithm, *VNU J. Sci.: Comput. Sci. Commun. Eng.*, 2014, vol. 30, no. 3, pp. 22–30.
- [17] Ionescu, C., Carreira, J., and Sminchisescu, C., Iterated second-order label sensitive pooling for 3D human pose estimation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2014, pp. 1661–1668.
- [18] Ionescu, C., Li, F., and Sminchisescu, C., Latent structured models for human pose estimation, *Proc. Int. Conf. on Computer Vision (ICCV)*, 2011, pp. 2220–2227.
- [19] Ionescu, C., Papava, D., Olaru, V., and Sminchisescu, C., Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments, *IEEE Trans. Pattern Anal. Mach. Intell.*, 2014, vol. 36, no. 7, pp. 1325–1339.
- [20] Johnson, S. and Everingham, M., Clustered pose and nonlinear appearance models for human pose estimation, *Proc. British Machine Vision Conf. (BMVC)*, 2010.
- [21] Johnson, S. and Everingham, M., Learning effective human pose estimation from inaccurate annotation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2011, pp. 1465–1472.
- [22] Kachurka, V., Rault, B., Muñoz, F.I.I., Roussel, D., Bonardi, F., Didier, J.-Y., Hadj-Abdelkader, H., Bouchafa, S., Alliez, P., and Robin, M., WeCo-SLAM: Wearable cooperative SLAM system for real-time indoor localization under challenging conditions, *IEEE Sens. J.*, 2021, vol. 22, no. 6, pp. 5122–5132.
- [23] Krizhevsky, A., Sutskever, I., and Hinton, G.E., ImageNet classification with deep convolutional neural networks, *Adv. Neural Inf. Process. Syst. (NIPS)*, 2012, vol. 25, pp. 1097–1105.
- [24] Long, J., Shelhamer, E., and Darrell, T., Fully convolutional networks for semantic segmentation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 3431–3440.
- [25] Mori, G. and Malik, J., Estimating human body configurations using shape context matching, *Proc. European Conf. on Computer Vision (ECCV)*, 2002, pp. 666–680.
- [26] Osadchy, M., LeCun, Y., and Miller, M.L., Synergistic face detection and pose estimation with energy-based models, *J. Mach. Learn. Res.*, 2007, vol. 8, pp. 1197–1215.
- [27] Pishchulin, L., Andriluka, M., Gehler, P., and Schiele, B., Poselet conditioned pictorial structures, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 588–595.
- [28] Popa, A.-I., Zanfır, M., and Sminchisescu, C., Deep multitask architecture for integrated 2D and 3D human sensing, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4714–4723.
- [29] Ramakrishna, V., Munoz, D., Hebert, M., Bagnell, J.A., and Sheikh, Y., Pose machines: Articulated pose estimation via inference machines, *Proc. European Conf. on Computer Vision (ECCV)*, 2014, pp. 33–47.

- [30] Ramanan, D., Learning to parse images of articulated bodies, *Adv. Neural Inf. Process. Syst. (NIPS)*, 2006, vol. 19, pp. 1129–1136.
- [31] Reyes-Ortiz, J., Anguita, D., Ghio, A., Oneto, L., and Parra, X., Human activity recognition using smartphones, *UCI Machine Learning Repository*, 2012.
- [32] Samsu, S., Maulidevi, N.U., and Aryan, P.R., Human action recognition using dynamic time warping, *Proc. 2011 Int. Conf. on Electrical Engineering and Informatics (ICEEI)*, IEEE, 2011, pp. 1–5.
- [33] Sapp, B. and Taskar, B., MODEC: Multimodal decomposable models for human pose estimation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 3674–3681.
- [34] Shafaei, A. and Little, J.J., Real-time human motion capture with multiple depth cameras, IEEE, 2012.
- [35] Shakhnarovich, G., Viola, P., and Darrell, T., Fast pose estimation with parameter-sensitive hashing, *Proc. Int. Conf. on Computer Vision (ICCV)*, 2003, pp. 750–757.
- [36] Sigal, L., Balan, A.O., and Black, M.J., HumanEva: Synchronized video and motion capture dataset and baseline algorithm for evaluation of articulated human motion, *Int. J. Comput. Vis.*, 2010, vol. 87, nos. 1–2, pp. 4–27.
- [37] Simonyan, K. and Zisserman, A., Very deep convolutional networks for large-scale image recognition, *CoRR*, 2015, abs/1409.1556.
- [38] Sun, H., Guanhuan, N., Zhiquan, Z., Zhongchao, H., and Zhihai, H., Automated work efficiency analysis for smart manufacturing using human pose tracking and temporal action localization, *J. Visual Commun. Image Represent.*, 2020, vol. 73, p. 102948.
- [39] Sun, Y., Wang, X., and Tang, X., Deep convolutional network cascade for facial point detection, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 3476–3483.
- [40] Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., and Rabinovich, A., Going deeper with convolutions, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1–9.
- [41] Szegedy, C., Toshev, A., and Erhan, D., Deep neural networks for object detection, *Adv. Neural Inf. Process. Syst. (NIPS)*, 2013, vol. 26, pp. 2553–2561.
- [42] Taylor, G.W., Fergus, R., Williams, G., Spiro, I., and Bregler, C., Pose-sensitive embedding by nonlinear NCA regression, *Adv. Neural Inf. Process. Syst. (NIPS)*, 2010, vol. 23, pp. 2280–2288.
- [43] Tian, Y., Zitnick, C.L., and Narasimhan, S.G., Exploring the spatial hierarchy of mixture models for human pose estimation, *Proc. European Conf. on Computer Vision (ECCV)*, 2012, pp. 256–269.
- [44] Vikhreva, N.A., *Dance Writing*, Moscow: The Moscow State Academy of Choreography, 2006.
- [45] Vikhreva, N.A., *History of Dance Writing*, Moscow: The Moscow State Academy of Choreography, 2014.
- [46] Wang, F. and Li, Y., Beyond physical connections: Tree models in human pose estimation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2013, pp. 596–603.
- [47] Wei, S.-E., Ramakrishna, V., Kanade, T., and Sheikh, Y., Convolutional pose machines, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 4724–4732.
- [48] Xia, F., Wang, P., Chen, X., and Yuille, A., Joint multi-person pose estimation and semantic part segmentation, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 6769–6778.

- [49] Xu, H., Bazavan, E.G., Zanfir, A., Freeman, B., Sukthankar, R., and Sminchisescu, C., GHUM & GHUML: Generative 3D human shape and articulated pose models, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 6183–6192.
- [50] Yang, Y. and Ramanan, D., Articulated pose estimation with flexible mixtures-of-parts, *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2011, pp. 1385–1392.



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